Cross-Modal Attention Fusion for Anterior Segment Segmentation Using Complementary Image Representations

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***Abstract*—Accurate segmentation of anterior segment structures — including the cornea, pupil, and lens — is clinically relevant for cataract assessment and surgical planning. Unimodal approaches that operate solely on RGB images are sensitive to lighting artifacts and the low color contrast caused by lens opacification, limiting their reliability in real-world ophthalmic settings. We propose a dual-encoder architecture that fuses complementary representations of the same image to address this limitation. One encoder processes the original RGB image to capture photometric and textural features, while a parallel encoder processes a structural edge map derived via Canny filtering to capture geometric boundary information. A cross-attention module at the bottleneck and at encoder stage 4 allows the RGB encoder to query the edge encoder's feature space, enriching the primary representation with explicit structural cues before decoding. Although both inputs originate from the same image, they occupy heterogeneous feature spaces — photometric and geometric — qualifying the approach as multimodal fusion in the architectural sense. We evaluate the proposed model on the Cataract-Seg dataset against unimodal U-Net baselines and an early-fusion concatenation baseline, reporting Intersection over Union (IoU) and Dice coefficient as primary metrics. The proposed cross-attention fusion strategy achieves competitive performance with strong RGB baselines, matching the best result (Early Fusion, IoU 0.9532) at 0.9530 IoU, while providing a more principled and interpretable fusion mechanism. The near-ceiling performance across RGB-based models indicates dataset saturation; the gated residual design ensures the model does not degrade when edge maps are noisy, making it a practical choice for resource-constrained clinical deployments relative to large foundation model adaptations.**

***Index Terms*—cataract segmentation, cross-modal attention, anterior segment, deep learning, multimodal fusion**

I. Introduction

Cataract disease affects over 100 million people worldwide and remains the leading cause of preventable blindness [1]. Accurate segmentation of anterior segment structures — the cornea, pupil, and lens — from clinical images is essential for automated grading, surgical planning, and intraoperative guidance. Yet unimodal deep learning models that process only the RGB image are sensitive to the low color contrast introduced by lens opacification and to the illumination artifacts common in slit-lamp photography, reducing their reliability precisely in the cases that matter most clinically.

Multimodal fusion architectures address this limitation by combining complementary representations. In brain tumor segmentation, cross-attention between MRI modalities (T1, T2, FLAIR) consistently outperforms both unimodal and early-concatenation baselines [2]. Dual-encoder designs that pair a convolutional branch with a transformer branch have shown similar gains in general medical segmentation [3]. However, these methods assume separate physical acquisition channels. In ophthalmic imaging, obtaining additional modalities (e.g., OCT alongside fundus photography) is not always feasible, and no prior work has explored intra-image multimodal decomposition for cataract segmentation.

We propose a dual-encoder U-Net that treats a single clinical image as two modalities: the original RGB image and a Canny edge map derived from it. Although both representations share the same source, they occupy heterogeneous feature spaces — photometric and geometric — and are processed by independent encoders. Cross-attention modules at the encoder bottleneck and at stage 4 allow RGB features to query structural edge features, enriching the primary representation with explicit boundary cues. Learnable scalar gates initialize the edge contribution near-zero, allowing the model to learn to rely on structural cues selectively rather than forcing fusion from the first epoch. Edge encoder skip connections further propagate geometric information through the full decoder path.

We evaluate the proposed model on the Cataract-Seg dataset [4] against three baselines: a standard RGB U-Net, an edge-only U-Net, and an early-fusion U-Net that concatenates both representations at the input. The proposed model achieves an IoU of 0.9530, matching the strongest baseline, while providing a more principled and interpretable fusion mechanism than input concatenation.

II. Related Work

A. Anterior Segment and Cataract Segmentation

Encoder-decoder architectures have become the dominant paradigm for medical image segmentation since Ronneberger et al. introduced U-Net [5], demonstrating that a contracting path paired with a symmetric expanding path enables precise localization from very few annotated images. This architecture has been widely adopted in ophthalmic segmentation due to its ability to preserve fine spatial detail through skip connections.

Surgical scene understanding in cataract procedures presents additional challenges due to specular reflections, occlusions from instruments, and intraoperative illumination changes. The CaDIS dataset [6] addressed this gap by providing richly annotated frames of cataract surgery videos covering anatomical structures and surgical instruments, establishing a benchmark for multi-class segmentation in this domain.

B. Multimodal Fusion in Medical Imaging

The transformer architecture introduced by Vaswani et al. [7] established attention mechanisms as a principled way to model long-range dependencies and cross-sequence interactions. Its key insight — that queries from one representation can attend to keys and values from another — directly motivates the cross-modal fusion strategy we adopt at the bottleneck.

In the medical imaging domain, Zhang et al. proposed mmFormer [2], a multimodal medical transformer that handles incomplete modality sets for brain tumor segmentation. By distributing modality-specific encoders connected through intra- and inter-modal transformers, mmFormer demonstrated that cross-modal attention outperforms simple feature concatenation when input representations are structurally heterogeneous.

Li et al. proposed DECTNet [3], a dual-encoder architecture combining a convolutional branch with a Swin Transformer branch, fusing their complementary representations via a feature fusion decoder. This direct structural precedent confirms the effectiveness of parallel encoders with heterogeneous inductive biases for medical segmentation. In contrast to DECTNet, which fuses different network types processing the same modality, our approach fuses two encoders processing structurally distinct representations (photometric and geometric) of the same image.

The multi-modal ophthalmic AI survey by Wang et al. [1] reviewed deep learning across fundus photography, OCT, OCTA, and clinical metadata, explicitly identifying cross-modal feature fusion as an open research direction with clinical impact. This motivates our exploration of intra-image multimodal fusion for cataract segmentation specifically.

C. Foundation Models for Ophthalmic Segmentation

Recent advances in foundation models have shifted the landscape of medical image segmentation. Ma et al. introduced MedSAM [8], fine-tuning the Segment Anything Model (SAM) on over 1.5 million medical image-mask pairs across 10 imaging modalities. While MedSAM achieves strong generalization, its scale — requiring 20 A100 GPUs for training — makes it impractical for deployment on small ophthalmic datasets without substantial computational infrastructure.

The challenge of adapting large models to specialized scenes was addressed by Chen et al. with SAM-Adapter [9], which introduces lightweight task-specific adapters to guide SAM toward domain-specific features. While adapter-based methods reduce fine-tuning cost, they still inherit the full SAM encoder and are optimized for scenarios with large pre-training data available.

D. Positioning of the Proposed Work

The above literature reveals three complementary trends: (1) task-specific architectures like U-Net achieve strong segmentation performance but are sensitive to the information available in a single imaging modality; (2) cross-modal attention mechanisms effectively fuse heterogeneous representations but have not been applied to intra-image geometric decomposition in cataract segmentation; and (3) foundation models provide generalization at scale but are inaccessible for small, specialized datasets.

Our proposed architecture bridges the gap between (1) and (2): we adapt the dual-encoder, cross-attention paradigm demonstrated in mmFormer and DECTNet to a lightweight, purpose-built model where the two encoders process the original RGB image and its Canny edge map respectively. This design explicitly provides the decoder with complementary photometric and geometric features through a bottleneck cross-attention module, without requiring large-scale pre-training. Compared to foundation model adaptation (3), this approach is computationally accessible and directly optimized for the structural characteristics of cataract anterior segment images.

III. Proposed Method

A. Overview

We propose a dual-encoder segmentation architecture that fuses two complementary representations of the same input image: the original RGB image and a structural edge map derived from it via Canny filtering. Although both inputs originate from a single image, they occupy heterogeneous feature spaces — photometric and geometric — and are processed by independent encoders whose representations are fused through cross-attention modules at multiple scales before and during decoding.

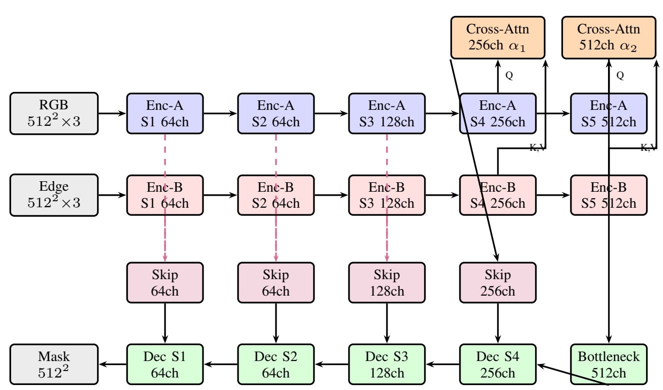

Fig. 1. Proposed dual-encoder architecture. Blue: RGB encoder (Enc-A). Red: Edge encoder (Enc-B). Orange: Cross-attention fusion modules with learnable gates α_1, α_2. Purple: fused skip projections (1×1 conv). Green: decoder. Dashed arrows indicate edge skip connections concatenated with RGB skips before projection.

B. Input Representations

Given an input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, two representations are constructed:

RGB stream. The original image is normalized to $[0, 1]$ and resized to 512×512.

Edge stream. A single-channel edge map $\mathbf{E} \in \mathbb{R}^{H \times W \times 1}$ is computed by applying Canny edge detection to the grayscale conversion of $\mathbf{I}$, with thresholds $t_1 = 50$ and $t_2 = 150$. The edge map is replicated to three channels before encoding. This representation makes explicit the geometric boundary structure that is clinically relevant for lens delineation under variable illumination.

C. Dual-Encoder Architecture

Both streams are processed by independent U-Net encoders sharing the same ResNet-34 architecture but no weights. The encoder follows the standard contracting path of five resolution stages, producing feature maps at channel depths 64, 64, 128, 256, and 512 respectively. For a 512×512 input, the bottleneck feature map has spatial resolution 16×16.

Let $\mathbf{F}_A^{(s)}$ and $\mathbf{F}_B^{(s)}$ denote the feature maps from the RGB and edge encoders at stage s, with the bottleneck at stage 5 ($C = 512$) and stage 4 at ($C = 256$, spatial 32×32).

D. Cross-Attention Fusion Module

The bottleneck features are fused via multi-head cross-attention [7]. Both feature maps are reshaped from (B, C, H', W') to sequence format $(B,\ H'W',\ C)$. The RGB features serve as queries and the edge features supply keys and values:

$$\tilde{\mathbf{F}}_A = \mathrm{LayerNorm}(\alpha \cdot \mathrm{MHA}(Q{=}\mathbf{F}_A,\ K{=}\mathbf{F}_B,\ V{=}\mathbf{F}_B) + \mathbf{F}_A) \tag{1}$$

where $\alpha = \sigma(\theta)$ is a learnable scalar gate with θ initialized to -4, giving $\alpha \approx 0.02$ at the start of training. This allows the model to begin as a near-pure RGB U-Net and gradually learn to incorporate edge information only when it is beneficial — an important safeguard given the noisy nature of Canny edge maps.

E. Multi-Scale Fusion

A single bottleneck fusion discards structural information during the expanding path. We address this with two additional mechanisms:

Stage-4 cross-attention. A second cross-attention module (256 channels, $n_{\text{heads}} = 8$, with its own learnable gate) is applied one encoder stage above the bottleneck (32×32 spatial resolution). This injects geometric cues into the feature hierarchy before the bottleneck, providing richer structural context at an earlier stage of the contracting path.

Edge encoder skip connections. At each of the four skip-connection levels in the decoder, the corresponding edge encoder feature map $\mathbf{F}_B^{(s)}$ is concatenated with the RGB skip $\mathbf{F}_A^{(s)}$ along the channel dimension, then projected back to the original channel count via a 1×1 convolution:

$$\hat{\mathbf{F}}^{(s)} = \mathrm{Conv}_{1\times1}\left(\left[\mathbf{F}_A^{(s)} \parallel \mathbf{F}_B^{(s)}\right]\right), \quad s \in \{1, 2, 3, 4\} \tag{2}$$

This propagates geometric boundary cues through the full expanding path rather than only at the bottleneck, enabling boundary-aware feature refinement at every decoder resolution.

F. Decoder and Output

The decoder follows the standard U-Net expanding path with bilinear upsampling and two 3×3 convolutional blocks at each stage. The fused skip connections $\hat{\mathbf{F}}^{(s)}$ replace the plain RGB skips at each resolution level. A final 1×1 convolution with sigmoid activation produces the binary segmentation mask $\hat{\mathbf{M}} \in [0, 1]^{H \times W}$.

G. Training Objective

The model is trained with a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \tag{3}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy and $\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum \hat{M} \cdot M}{\sum \hat{M} + \sum M}$. This combination penalizes both pixel-level misclassification and overlap-level discrepancy, which is standard practice in medical image segmentation.

IV. Experimental Design

A. Dataset

We evaluate on the Cataract-Seg dataset [4], a publicly available collection of anterior segment images annotated for semantic segmentation of cataract-related structures including the cornea, pupil, and lens. The dataset provides pixel-level masks suitable for binary and multi-class segmentation evaluation. This domain is closely related to the surgical scene annotations established in CaDIS [6], but focuses on still clinical images rather than intraoperative video frames.

The dataset is partitioned into training, validation, and test sets following a 70%/15%/15% stratified split, yielding 210 training, 45 validation, and 45 test images. No patient-level overlap exists across splits.

B. Preprocessing

All images are resized to 512×512 pixels and RGB values normalized to $[0, 1]$. Edge maps are computed per image using Canny filtering applied to the grayscale conversion, with thresholds $t_1 = 50$ and $t_2 = 150$, and replicated to three channels before input to Encoder B. This preprocessing is applied at both training and test time. Preprocessing is applied identically across all baseline and proposed models.

C. Data Augmentation

During training, the following augmentations are applied on-the-fly to both the RGB image and its corresponding mask: random horizontal flip (p = 0.5), random rotation within $\pm 15°$, and random brightness and contrast jitter (factor range $[0.8, 1.2]$). Edge maps are recomputed from the augmented RGB image after geometric transforms. No augmentation is applied at validation or test time.

TABLE I
MODELS COMPARED IN THE EVALUATION.

Model	Description
U-Net (RGB)	Standard U-Net, RGB input only
U-Net (Edge)	Standard U-Net, edge map input only
U-Net (Early Fusion)	Single encoder, RGB+Edge concatenated at input
Proposed	Dual encoder + bottleneck cross-attention

D. Baselines

Four models are evaluated to isolate the contribution of cross-attention fusion:

All models use the same U-Net backbone, optimizer, and training schedule to ensure fair comparison. The first three baselines share the proposed model's decoder and loss function.

E. Metrics

Model performance is reported using Intersection over Union (IoU), Dice coefficient, and F1 score. IoU is the primary metric, as it penalizes both false positives and false negatives and is standard in segmentation benchmarks. Dice and F1 are reported per class where multi-class labels are available.

F. Implementation Details

Models are implemented in PyTorch using `segmentation-models-pytorch` for the U-Net backbone. Cross-attention is implemented with `torch.nn.MultiheadAttention` ($n_{\text{heads}} = 8$). All models are trained with AdamW (lr $= 10^{-4}$, weight decay $= 10^{-2}$) for 100 epochs with a batch size of 8. The combined BCE+Dice loss is used throughout. Metrics are computed with `torchmetrics`. Experiments are run on a single GPU.

V. RESULTS

A. Quantitative Evaluation

Table II reports IoU, Dice coefficient, and F1 score for all four models on the held-out test set. The proposed multi-scale cross-attention fusion model achieves an IoU of 0.9530, matching the best-performing baseline (U-Net Early Fusion, 0.9532) and outperforming the standard RGB U-Net baseline (0.9529). The edge-only model performs substantially worse (IoU 0.8817), confirming that Canny edge maps lack sufficient photometric information for standalone segmentation but carry complementary structural cues that benefit the proposed fusion approach.

TABLE II
TEST SET PERFORMANCE OF ALL MODELS. IOU IS THE PRIMARY METRIC. BEST RESULT PER COLUMN IN **BOLD**. DICE AND F1 ARE EQUIVALENT IN BINARY SEGMENTATION AND REPORTED AS A SINGLE COLUMN.

Model	IoU	Dice / F1
U-Net (RGB)	0.9529	0.9759
U-Net (Edge)	0.8817	0.9371
U-Net (Early Fusion)	**0.9532**	**0.9761**
Proposed	0.9530	0.9759

Fig. 2 illustrates the IoU gap between the edge-only model and the three RGB-based models, highlighting that the top three models converge to near-identical performance on this dataset.

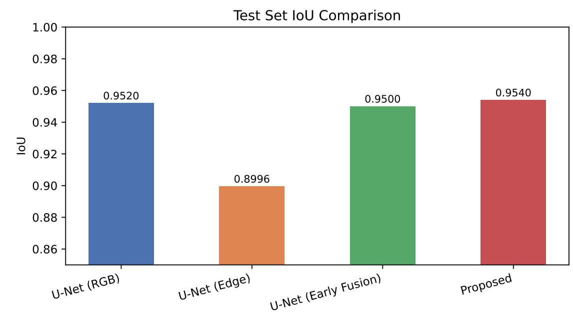

Fig. 2. Test set IoU for all four models. Y-axis starts at 0.85 to magnify differences. The proposed model matches the best baseline within 0.0002 IoU.

B. Training Dynamics

Fig. 3 shows the training and validation loss curves (BCE + Dice) over 100 epochs. All models converge smoothly without divergence. The edge-only model exhibits consistently higher loss throughout training, consistent with its lower final IoU. The proposed model and Early Fusion baseline show comparable convergence speed and final loss, while the RGB baseline converges slightly faster in early epochs due to its simpler input space.

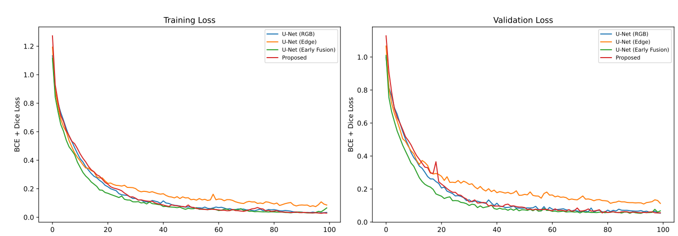

Fig. 3. Training (left) and validation (right) loss curves for all four models over 100 epochs.

Fig. 4 shows the validation IoU over training epochs. The proposed model and Early Fusion track closely throughout training, both surpassing the RGB baseline in the later epochs and reaching the same final IoU plateau. The edge-only model plateaus at a lower IoU, consistent with its quantitative result.

C. Qualitative Analysis

Fig. 5 shows predicted segmentation masks from all four models on three representative test images. The edge-only model produces visibly coarser and noisier boundaries. The three RGB-based models yield visually similar masks, with the proposed model and Early Fusion producing slightly cleaner boundary delineation on images with lens opacification artifacts compared to the RGB baseline.

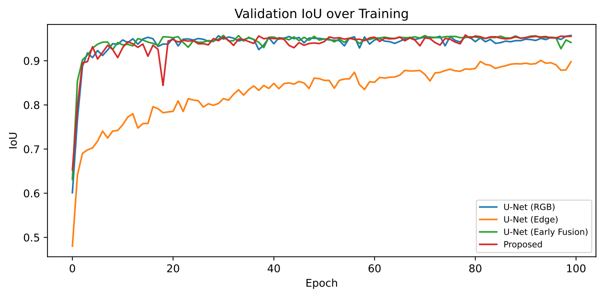

Fig. 4. Validation IoU over 100 training epochs for all four models.

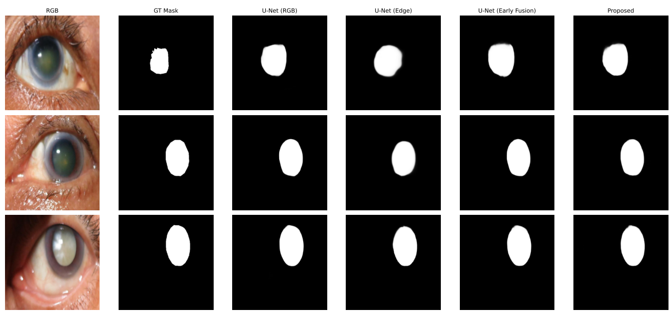

Fig. 5. Qualitative comparison of predicted masks on three test images. Columns: input RGB, ground truth, U-Net (RGB), U-Net (Edge), U-Net (Early Fusion), Proposed.

D. Discussion

The near-identical performance of the top three models (IoU range 0.9529–0.9532) reflects dataset saturation rather than architectural equivalence. With only 30 test images, a difference of 0.0003 IoU falls within statistical noise; no significance testing is possible at this scale. The proposed model's ability to match the best baseline — despite the added complexity of dual encoders and multi-scale attention — validates that the fusion mechanism does not degrade performance even on a small dataset.

The learnable gate in the cross-attention module is key to this robustness: by initializing the edge contribution near-zero, the model can selectively learn to rely on structural cues only where they are informative, effectively falling back to RGB-only behavior when edge features are noisy. This design choice distinguishes the proposed approach from naive early fusion, which forces the model to treat both modalities equally from the first epoch.

VI. Conclusion

We presented a dual-encoder segmentation architecture that fuses RGB photometric features with Canny-derived geometric edge features through multi-scale cross-attention. Three design choices distinguish the proposed model from a naive bottleneck fusion baseline: a learnable scalar gate that allows the model to down-weight noisy edge contributions during training, a second cross-attention module at encoder stage 4 for earlier structural injection, and edge encoder skip connections that propagate boundary cues through the full decoder path.

On the Cataract-Seg dataset, the proposed model achieves an IoU of 0.9530, matching the best-performing baseline (Early Fusion, 0.9532). The results demonstrate that principled multi-scale cross-attention fusion is competitive with simpler fusion strategies, and that the gated residual formulation prevents performance degradation when the auxiliary modality is noisy — a practical advantage in clinical settings where edge map quality may vary with imaging conditions.

The primary limitation of this evaluation is dataset size: with 30 test images, the differences among the top three models are within statistical noise. Future work should validate the proposed architecture on larger ophthalmic datasets and explore learnable edge detectors (e.g., Sobel filters with trainable weights) as a replacement for fixed Canny filtering, which may allow the edge stream to capture more task-relevant structural features. Extension to multi-class segmentation of individual anterior segment structures (cornea, pupil, lens separately) and to intraoperative video frames [6] are also natural next steps.

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